Math 477 Homework 6

Spring 2026
Rutgers University

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Exercise 1:

Let X denote number of minutes past 10 o'clock that the bus takes to arrive. Then, $X \sim U(0, 30)$, with PDF:

$$f(x) = \begin{cases} \frac{1}{30}, & 0 < x < 30 \\ 0, & \text{otherwise.} \end{cases}$$

Since you got here 10 o'clock, if the bus gets here after 10:10, you have waited more than 10 minutes. That is,

$$\begin{aligned} P(\text{wait} > 10 \text{ minutes}) &= P(10 < X < 30) \\ &= \int_{10}^{30} \frac{1}{30} dx \\ &= \frac{20}{30} \\ &= \frac{2}{3}. \end{aligned}$$

Exercise 2:

Let $X \sim N(100, 15)$. We standardize using the transformation:

$$\begin{aligned} Z &= \frac{X - \mu}{\sigma} \\ &= \frac{X - 100}{15} \end{aligned}$$

(a)

$$\begin{aligned} P(X > 125) &= P(Z > \frac{125 - 100}{15}) \\ &= P(Z > \frac{25}{15}) \\ &\approx P(Z > 1.67) \\ &\approx 1 - 0.9525 \\ &= 0.0475. \end{aligned}$$

(b)

$$\begin{aligned} P(90 < X < 110) &= P(\frac{-10}{15} < Z < \frac{10}{15}) \\ &\approx P(-.67 < Z < .67) \\ &= \Phi(.67) - \Phi(-.67) \\ &\approx 0.7486 - 0.2514 \\ &= 0.4972. \end{aligned}$$

Exercise 3:

Integrating the entire PDF must be 1. That is,

$$\begin{aligned} 1 &= \int_{-\infty}^{\infty} f(x)dx \\ &= \int_0^1 a + bx^2 dx \\ &= [ax + \frac{1}{3}bx^3]_0^1 \\ &= a + \frac{b}{3}. \end{aligned}$$

Then, using expected value formula, we have

$$\begin{aligned} \frac{3}{5} &= \int_{-\infty}^{\infty} x \cdot f(x)dx \\ &= \int_0^1 ax + bx^3 dx \\ &= [\frac{1}{2}ax^2 + \frac{1}{4}bx^4]_0^1 \\ &= \frac{a}{2} + \frac{b}{4}. \end{aligned}$$

Scaling the two equations, we have the following system:

$$\begin{aligned} 3 &= 3a + b \\ \frac{12}{5} &= 2a + b. \end{aligned}$$

Subtracting the first from the second, we have

$$\begin{aligned} a &= 3 - \frac{12}{5} \\ &= \frac{3}{5}. \end{aligned}$$

Substituting a into the first equation, we have

$$\begin{aligned} \frac{3(3)}{5} + b &= 3 \\ b &= 3 - \frac{9}{5} \\ &= \frac{6}{5}. \end{aligned}$$

Therefore, we have $a = \frac{3}{5}$ and $b = \frac{6}{5}$.

Exercise 4:

Since $P(X > 9) = 0.2$, we have that $P(X < 9) = 0.8$. Thus, the Z-score is approximately 0.84. That is,

$$\begin{aligned} Z &= \frac{X - \mu}{\sigma} \\ 0.84 &= \frac{4}{\sigma} \\ \sigma &= \frac{4}{0.84} \\ &\approx 4.76. \end{aligned}$$

Since $Var(X) = \sigma^2$, we have

$$\begin{aligned} Var(X) &\approx 4.76^2 \\ &\approx 22.66 \end{aligned}$$

Exercise 5:

Suppose $X \sim N(40, 7)$. Then, the Z-score for 95th percentile is approximately 1.645. Plugging into our formula, we have

$$\begin{aligned} Z &= \frac{X - \mu}{\sigma} \\ 1.645 &= \frac{X - 40}{7} \\ X &= 51.515. \end{aligned}$$

Thus, we need to leave 51.515 minutes before 1:00 pm or around 12:08 pm.

Exercise 6:

Suppose $X \sim N(40, 4)$.

$$\begin{aligned} P(X > 50) &= P(Z > 2.5) \\ &\approx 0.0062 \end{aligned}$$

Then,

$$P(\text{First 10 years} \leq 50) = (0.9938)^{10}$$

$$P \approx 0.9396$$

Exercise 7:

Suppose the minimum bid (in thousands of dollars) of the other participating companies can be modeled as $X \sim U(70, 140)$, with PDF:

$$f(x) = \begin{cases} \frac{1}{70}, & 70 < x < 140 \\ 0, & \text{otherwise.} \end{cases}$$

Let y represent our chosen bid. Then, our profit is modeled by

$$g(X, y) = \begin{cases} y - 100, & X \geq y \\ 0, & y < X \end{cases}$$

So the expected value of our profit is

$$\begin{aligned} E[g(X, y)] &= \int_{-\infty}^{\infty} g(X, y) f(x) dx \\ &= \int_{y}^{140} \frac{y - 100}{70} dx \\ &= \frac{y - 100}{70} \int_{y}^{140} dx \\ &= \frac{y - 100(140 - y)}{70} \\ &= \frac{1}{70}(-y^2 + 240y - 14000). \end{aligned}$$

Setting to zero to find max, we have

$$\begin{aligned} 0 &= y^2 - 240y + 14000 \\ &= \end{aligned}$$